

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Izzati Ayub
Program/Batch	B083-Networkwalks
Date	20 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM4 (theHarvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

Performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f** and **DNSRecon**. Each tool was used to collect a different type of information about the target.

Used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

Used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.1** and **WP Download Manager 3.3.58**, along with other information exposed by the website.

Using **Nslookup**, resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

Used **Curl** with the **-I** option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint `/wp-json/`.

Used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

Domain

↓

networkwalks.com

↓

192.232.216.135

├— Web service

└— Mail service

DNS provider/infrastructure

├— ns6135.hostgator.com

└— ns6136.hostgator.com

Email infrastructure

├— mail.networkwalks.com

├— SPF policy

└— cPanel autodiscovery

4.3 Footprinting with theHarvester

TheHarvester was used to perform passive footprinting against `microsoft.com` using publicly available OSINT sources. An initial Baidu search identified five hosts, including `appsource.microsoft.com`, `code.msdn.microsoft.com`, and `officecdn.microsoft.com`, but returned no emails or IP addresses.

A broader search using multiple sources identified **157 IP addresses, 3 email addresses, 12 ASNs, 5 interesting URLs, and 9,961 hosts**. The discovered email addresses including:

- dotnet-docker-bot@microsoft.com
- opencode@microsoft.com
- secure@microsoft.com

The large number of hosts demonstrates the extent of the publicly visible attack surface associated with the domain. Interesting URLs also revealed Microsoft authentication endpoints, providing information about authentication infrastructure and application integrations. However, several data sources could

not be queried because API keys were unavailable; therefore, the results represent a partial collection of publicly available information rather than a complete enumeration. The findings demonstrate how passive OSINT can provide useful information for reconnaissance without directly interacting with or exploiting the target infrastructure.

4.3 Network Scanning with Zenmap

For network scanning **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows `ipconfig` command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

192.168.100.0/24

The results provided identified five live hosts:

- 192.168.100.100 - D4:C8:B0:B6:25:D3
- 192.168.100.103 - F2:16:ED:F8:9A:85
- 192.168.100.104 - C6:B8:C5:5D:3C:28
- 192.168.100.105 - 0C:F0:7B:6F:F8:6F
- 192.168.100.254 - 0C-F0-7B-6F-F8-6F

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence Observation /	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
6	Multiple live hosts visible on local network	Zenmap identified five live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

- 1. Review publicly exposed technology information**

Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**
CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance, and network scanning.

For the footprinting activities, I used seven Kali Linux tools to gather information about target domains. I used **WHOIS** to obtain domain information, **WhatWeb** to identify web technologies, **Nslookup** to perform DNS lookups, **cURL** to inspect HTTP headers, **Wafw00f** to identify web application firewalls, **DNSRecon** to gather DNS information, and **theHarvester** to collect publicly available information such as email addresses, IP addresses, hosts, ASNs, and URLs.

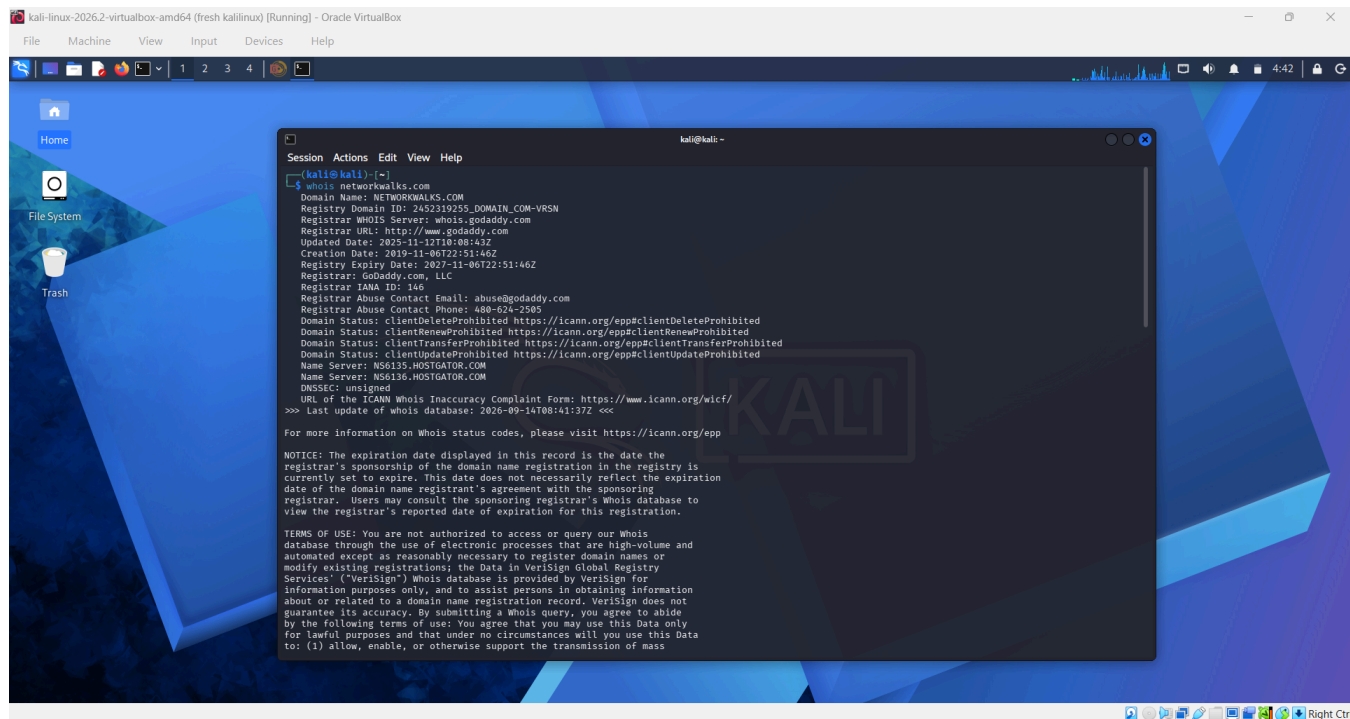
For the network scanning activity, I used **Zenmap** to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

These activities helped me understand that information gathering is an important part of cybersecurity. Before attempting to exploit a system, security professionals can gather useful information by analyzing publicly available data and network responses.

I also learned the importance of documenting technical findings clearly. A good cybersecurity report should explain what was performed, what was discovered, what the findings mean, and what risks or security concerns they may present.

Finally, I learned that reconnaissance and scanning should always be performed within an authorized scope. All activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

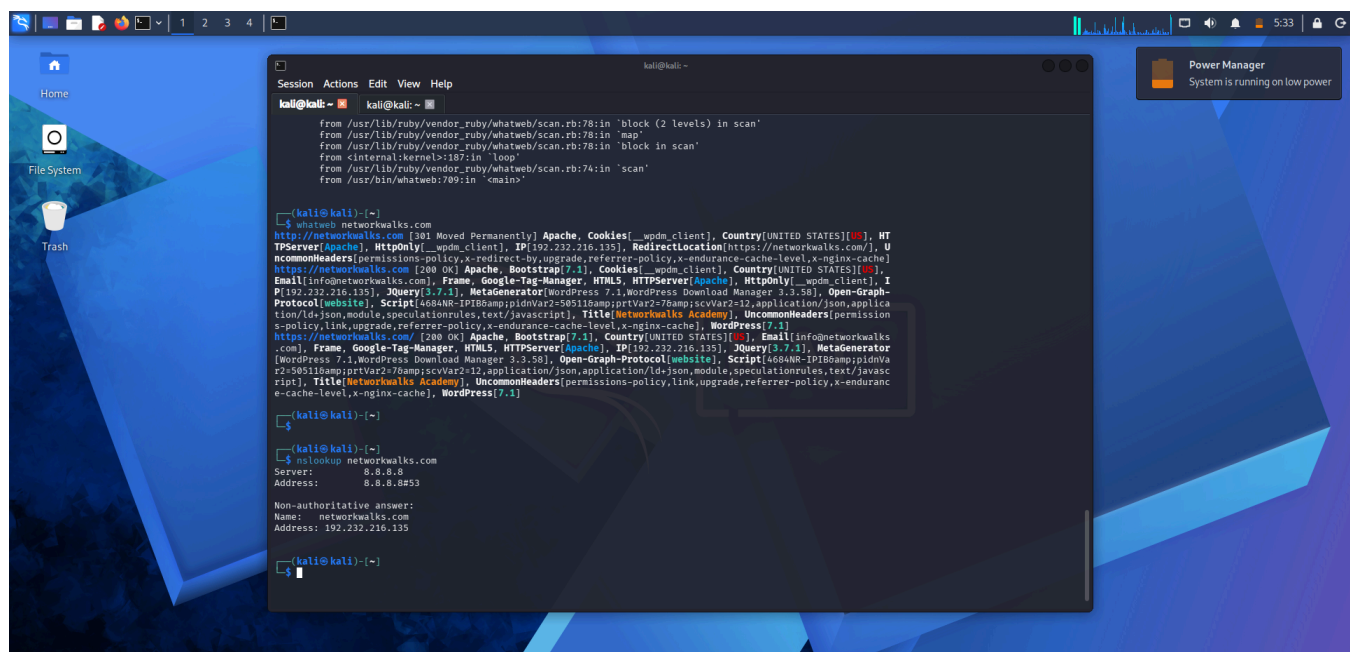


```
kali@kali:~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-17T10:08:45Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-14T08:41:37Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

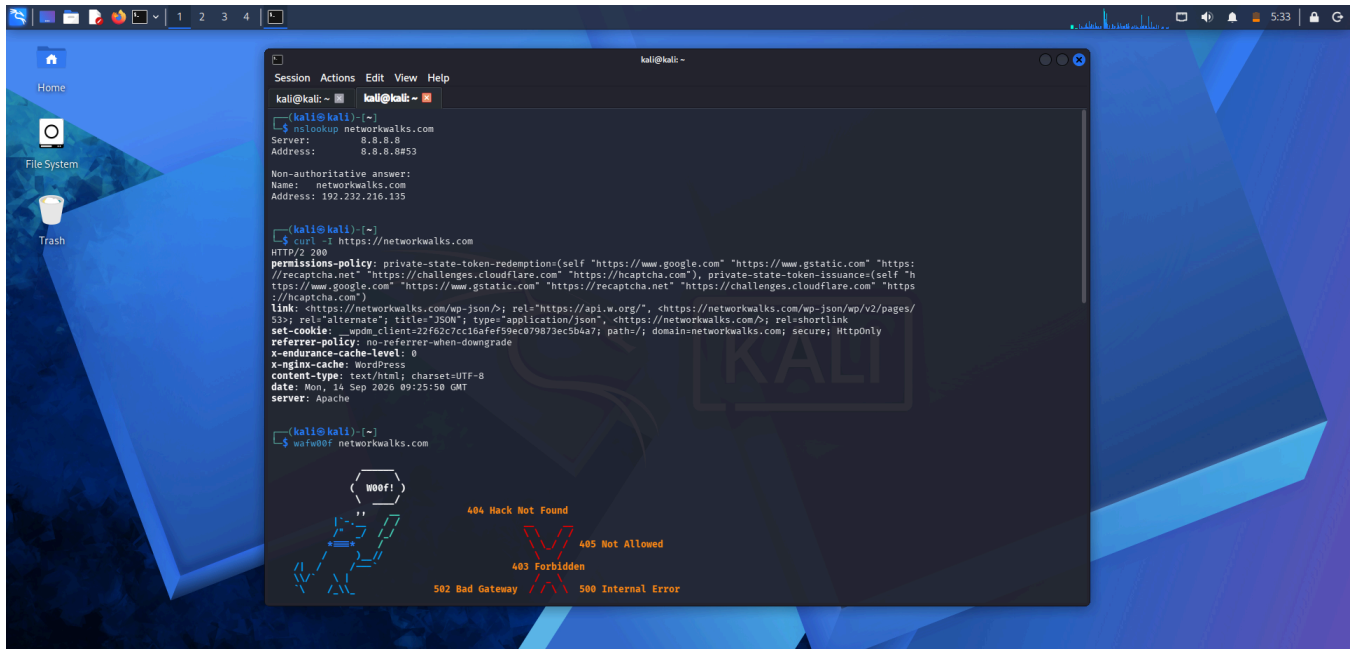
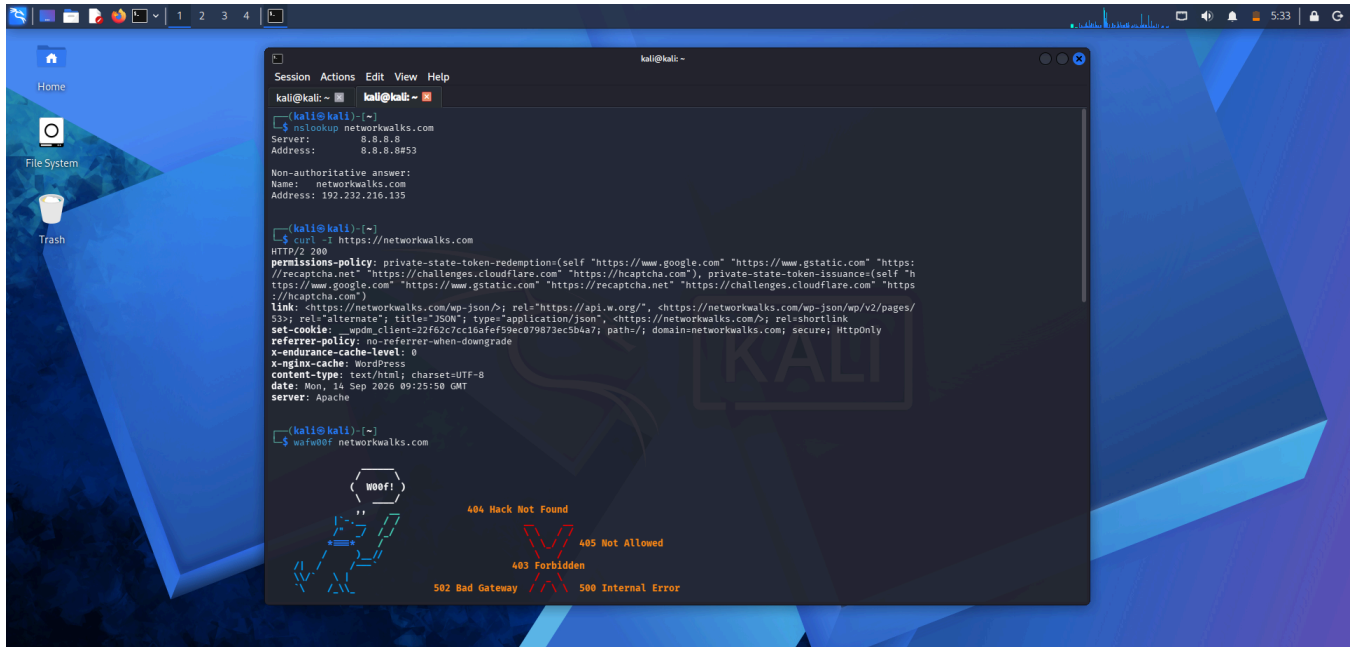
TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
```

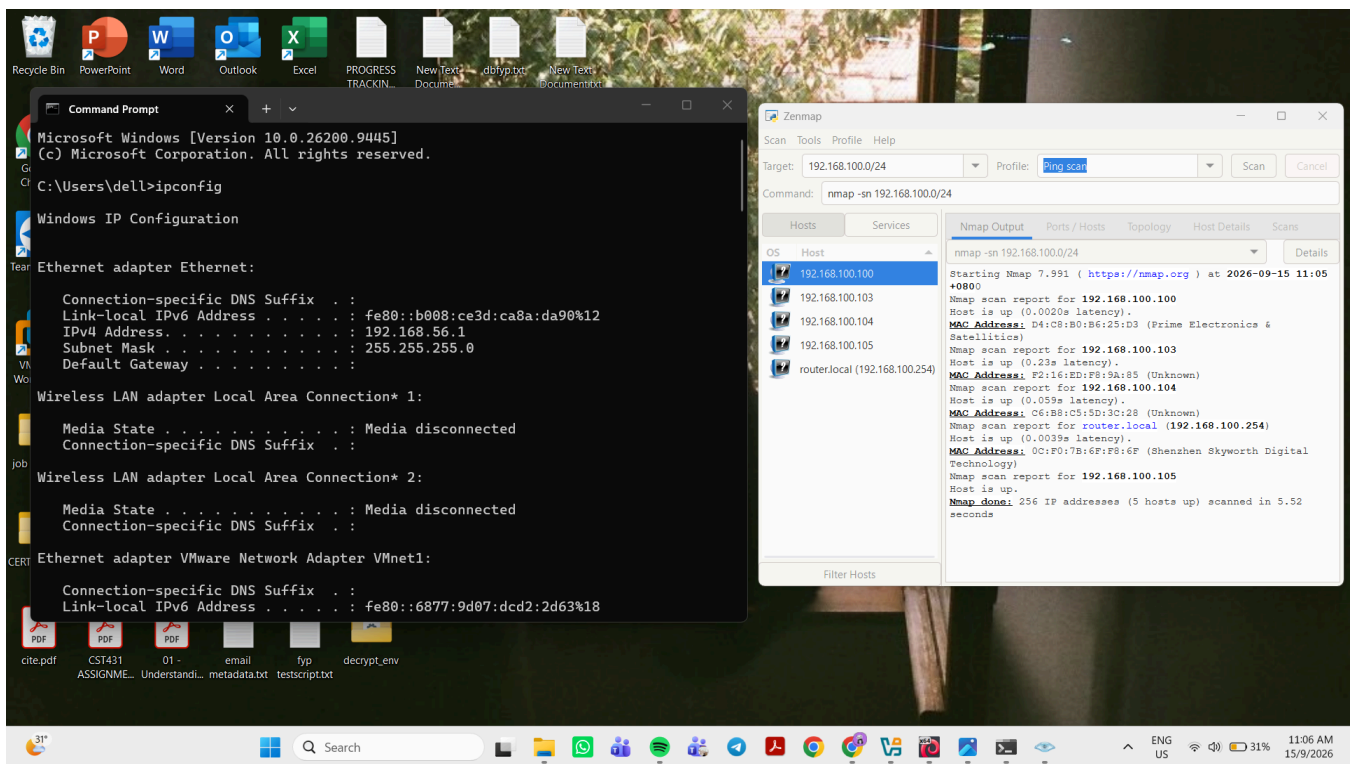
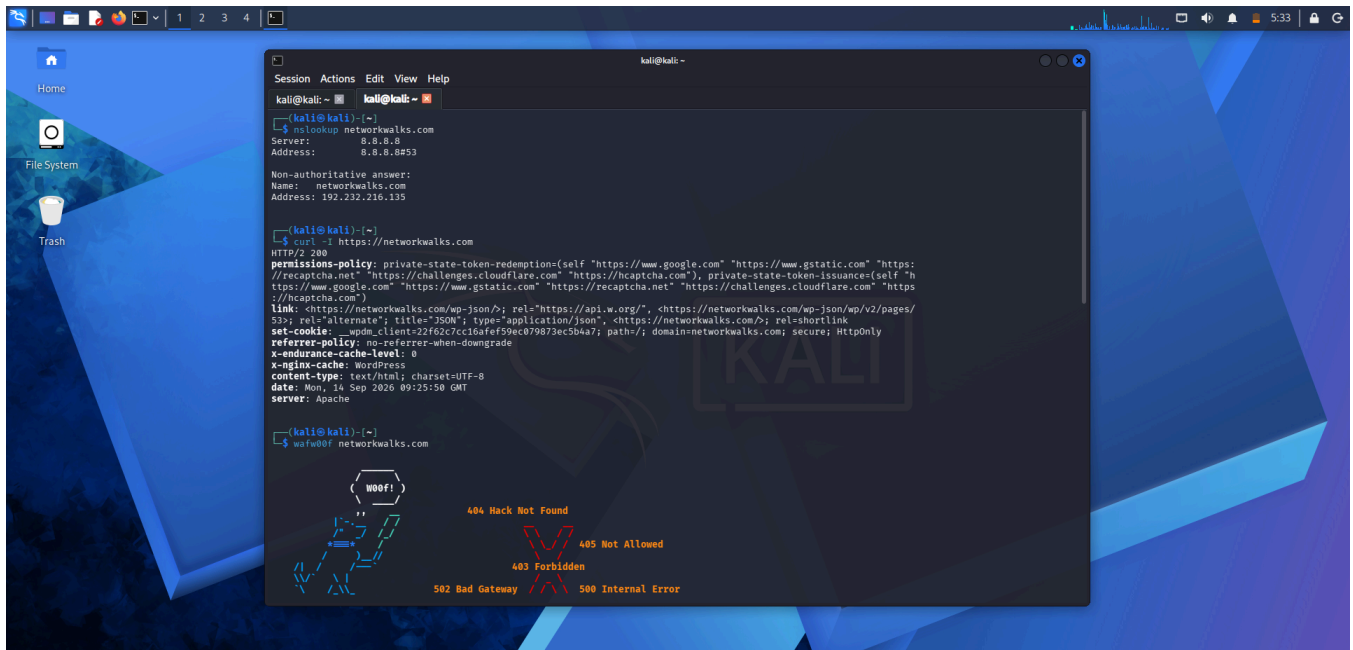


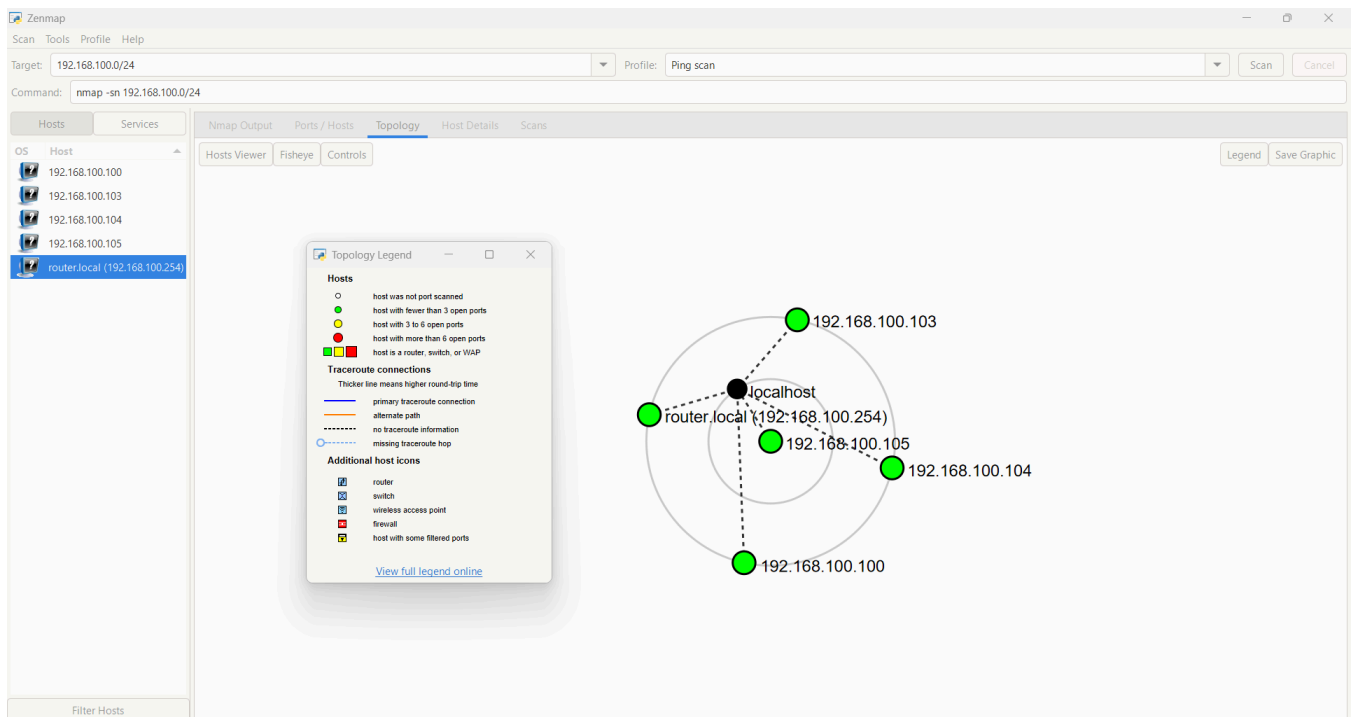
```
kali@kali:~$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpdm_client], Country[UNITED STATES][40], HT
TPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], U
ncommonHeaders[permissions-policy,x-redirect-by,upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[_wpdm_client], Country[UNITED STATES][1],
Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], I
P[192.232.216.135], JQuery[3.7.1], MetaGenerator[wordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-
Protocol[website], Script[468ANR-IPID&pidnVa2=5051&pidnVa2=7&pidnVa2=2&application/json,application/json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permission
s-policy,link,upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][1], Email[info@networkwalks
.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator
[wordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[468ANR-IPID&pidnVa
2=5051&pidnVa2=7&pidnVa2=2&application/json,application/json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referer-policy,x-enduranc
e-cache-level,x-nginx-cache], WordPress[7.1]

kali@kali:~$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```







```
fyp tes

File Edit View H1 ...

192.168.100.100 - D4:C8:B0:B6:25:D3
192.168.100.103 - F2:16:ED:F8:9A:85
192.168.100.104 - C6:B8:C5:5D:3C:28
192.168.100.105 - 0C:F0:7B:6F:F8:6F
192.168.100.254 - 0C-F0-7B-6F-F8-6F

Ln 10, Col 1 | 184 charact | Plain text | 100% | Windows ( | UTF-8
```

-End-

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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub